

Entropy Concavity for All Three-Bit Overlap Bernoulli Distributions

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Abstract

Let B_1, B_2, B_3 be independent Bernoulli variables with $\mathbb{P}(B_i = 1) = 1 - p$, and let u, v be arbitrary positive integers. We prove that the Shannon entropy of $1 + B_1 + uB_2 + vB_3$ is concave in p on $[0, 1]$. Every collision of subset sums of $\{1, u, v\}$ is classified, giving five exact entropy profiles. Each profile has the form $3b(p) - \delta p(1 - p)b(p) - cp(1 - p)$, where b is binary entropy; a single second-derivative estimate proves strict concavity on $(0, 1)$, and continuity handles the endpoints. This settles exactly the three-bit layer of the recursive overlap-Bernoulli entropy conjecture and makes no assertion for larger layers.

1 Statement and context

We use Shannon entropy [1], take natural logarithms, and adopt the convention $0 \log 0 = 0$. Neunhäuserer introduced recursively overlapping Bernoulli distributions and conjectured entropy concavity at every layer [2]. The cited source proves the binomial and completely non-overlapping regimes and reports symbolic evidence for further families. Here we address exactly its complete three-bit layer, including all overlap patterns.

For $p \in [0, 1]$, put $q = 1 - p$ and let B_1, B_2, B_3 be independent with $\mathbb{P}(B_i = 0) = p$ and $\mathbb{P}(B_i = 1) = q$. Given $u, v \in \mathbb{Z}_{>0}$, define

$$X_{u,v}(p) = 1 + B_1 + uB_2 + vB_3, \quad h_{u,v}(p) = H(X_{u,v}(p)).$$

The additive constant does not affect entropy. Every increasing prefix $2 = a_1 < a_2 < a_3$ corresponds to $u = a_2 - a_1$ and $v = a_3 - a_2$, and conversely every positive pair (u, v) arises in this way.

Theorem 1. *For every $(u, v) \in \mathbb{Z}_{>0}^2$, the map $p \mapsto h_{u,v}(p)$ is concave on $[0, 1]$. More precisely, its second derivative is strictly negative on $(0, 1)$.*

The case $(u, v) = (1, 1)$ is a three-trial binomial entropy profile, so it also follows from the Shepp–Olkin theorem [3]. We include it in the elementary argument below in order to treat every collision profile uniformly.

2 All collision profiles

Write

$$b(p) = -p \log p - q \log q, \quad R = pq,$$

and, on the open interval, set

$$A = p^3, \quad B = p^2q = pR, \quad C = pq^2 = qR, \quad D = q^3.$$

Lemma 2 (Exhaustive collision classification). *The multiset of positive atom probabilities of $X_{u,v}(p)$, together with its entropy, is exactly one of the following five rows.*

Parameters	Atom probabilities	Entropy profile
$(u, v) = (1, 1)$	$A, 3B, 3C, D$	$3b - 3R \log 3$
$(u, v) = (1, 2) \text{ or } (2, 1)$	$A, 2B, R, 2C, D$	$3b - Rb - 2R \log 2$
$u = 1, v \geq 3; v = 1, u \geq 3; \text{ or } u = v \geq 2$	$A, B, 2B, C, 2C, D$	$3b - 2R \log 2$
$ u - v = 1 \text{ and } \min\{u, v\} \geq 2$	A, B, B, R, C, C, D	$3b - Rb$
Every remaining pair	A, B, B, B, C, C, C, D	$3b$

Proof. Ignore the additive shift by 1. The support point associated with a bit vector $\varepsilon \in \{0, 1\}^3$ is the subset sum

$$s(\varepsilon) = \varepsilon_1 + u\varepsilon_2 + v\varepsilon_3.$$

If two distinct subset sums coincide, cancel their common summands. The two remaining subsets of $\{1, u, v\}$ are disjoint and nonempty. Because there are only three positive weights, the resulting equality either equates two weights or equates one weight with the other two. Thus a collision can occur only under one of

$$u = 1, \quad v = 1, \quad u = v, \quad v = u + 1, \quad u = v + 1;$$

the remaining formal relation $1 = u + v$ is impossible for positive integers. Conversely, each displayed relation causes a collision. Resolving their intersections gives $(1, 1)$, the two exceptional pairs $(1, 2), (2, 1)$, the three one-relation families in the third row, the consecutive families in the fourth row, and the collision-free complement. Directly grouping the eight bit vectors gives the atom multisets displayed in the table.

Before merging equal support points, independence gives entropy $3b$. Merging k equal atoms, each of mass a , lowers entropy by $ka \log k$. The only merge involving unequal bit-vector probabilities uses $B + C = R$; since $B = pR$ and $C = qR$,

$$[-B \log B - C \log C] - [-R \log R] = Rb.$$

These two identities yield all five entropy profiles in the final column. □

3 A uniform curvature bound

Lemma 2 reduces every case to

$$F_{\delta,c}(p) = 3b(p) - \delta Rb(p) - cR, \tag{1}$$

where, in table order,

$$(\delta, c) = (0, 3 \log 3), (1, 2 \log 2), (0, 2 \log 2), (1, 0), (0, 0).$$

Proposition 3. *Every profile in (1) satisfies $F''_{\delta,c}(p) < 0$ for $0 < p < 1$.*

Proof. On $(0, 1)$, direct differentiation gives

$$b'(p) = \log(q/p), \quad b''(p) = -\frac{1}{R}, \tag{2}$$

and, using $R' = 1 - 2p$ and $R'' = -2$,

$$(Rb)'' = -2b + 2(1 - 2p) \log(q/p) - 1. \tag{3}$$

If $\delta = 0$, equations (1)–(2) give

$$F''_{0,c} = -\frac{3}{R} + 2c.$$

Because $R \leq 1/4$, this is negative for $c = 3 \log 3$, $2 \log 2$, and 0 ; indeed $\log 3 < 2$ and $\log 2 < 1$.

Suppose $\delta = 1$. From (3),

$$F''_{1,c} = -\frac{3}{R} + 2b - 2(1 - 2p) \log(q/p) + 1 + 2c. \quad (4)$$

The binary entropy satisfies $b \leq \log 2$, since $b' = \log(q/p)$ changes sign only at $p = 1/2$. Moreover, $(1 - 2p) \log(q/p) \geq 0$, because its two factors have the same sign. Hence

$$F''_{1,c} \leq -11 + 2 \log 2 + 2c.$$

For $c = 0$ the right side is less than -9 , while for $c = 2 \log 2$ it is less than -5 . Thus both remaining profiles are strictly concave on the open interval. $\square$

Proof of Theorem 1. Proposition 3 proves concavity on every compact subinterval of $(0, 1)$. Every atom probability is a polynomial in p , and $x \log x \rightarrow 0$ as $x \downarrow 0$, so every profile in Lemma 2 extends continuously to $[0, 1]$. Apply the interior concavity inequality to inward approximations of any two endpoint arguments and pass to the limit. This proves concavity on the closed interval, including $p = 0$ and $p = 1$. $\square$

4 Scope and reproducibility

This proof resolves exactly the three-bit layer. Relative to [2], rows 1 and 5 are the binomial and non-overlapping extremes, while rows 2–4 are the intervening collisions. We prove no result for four or more bits and make no priority claim beyond this comparison.

One Python/SymPy script checks the five profile and derivative identities; a second uses integer arithmetic to reconstruct the partitions and checks both representatives and all 10,000 pairs with $1 \leq u, v \leq 100$. The computation is corroborative; Lemma 2 proves the universal claim. A comparable separation of proof from exact checks appears in [4, 5]; neither reference supports the present theorem.

An AI language model assisted in organizing the cases, checking derivations, and drafting. The theorem itself is proved in the text.

References

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